



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
28-03-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

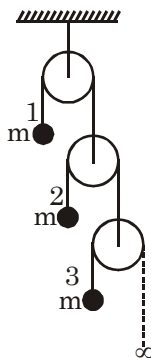
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

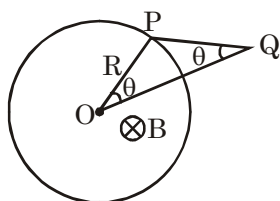
1. An Atwood's machine having infinite number of masses connected with ideal strings as shown in the figure. All the masses are identical and equal to m . All the pulleys are ideal. Then choose the **CORRECT** option(s).



- (A) Acceleration of mass-1 is $\frac{g}{2}$
 (B) Ratio of acceleration of mass-1 and mass-2 is 2
 (C) Total work done by tension is zero
 (D) Acceleration of mass-1 can't be determined

Space for Rough Work

2. In a circular region centered at O and having radius R, magnetic field increases at a constant rate a_0 T/sec. A wire frame OPQ having resistance per unit length λ Ω /m is placed as shown in the figure. Choose the **CORRECT** option(s).



(A) $V_P = V_Q$ for any θ as there is no induced emf across OP.

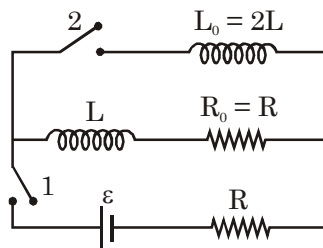
(B) $V_O - V_Q = \frac{a_0 \pi R^2}{8(\sqrt{2} + 1)}$ for $\theta = 45^\circ$

(C) $V_O - V_Q = \frac{a_0 \pi R^2}{2(\sqrt{2} - 1)}$ for $\theta = 45^\circ$

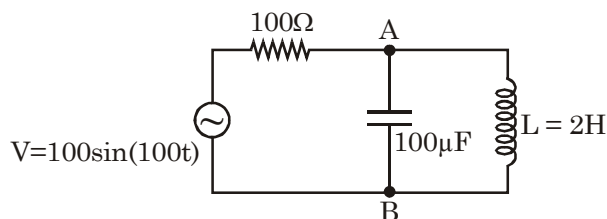
(D) $V_P - V_Q = \frac{a_0 \pi R^2}{8\sqrt{2}}$ for $\theta = 45^\circ$

Space for Rough Work

3. In the circuit shown, initially only switch-1 is closed for a long time. Then switch-2 is also closed. Choose the **CORRECT** option(s).



- (A) Current through the inductor ' L_0 ' in the steady state after closing the switch-2 will be $\frac{\varepsilon}{R}$.
- (B) Current through the inductor ' L_0 ' in the steady state after closing the switch-2 will be $\frac{\varepsilon}{2R}$.
- (C) The total charge $\left(\frac{5\varepsilon L}{2R^2}\right)$ will flow through the resistor ' R_0 ' after closing the switch-2.
- (D) The total charge $\left(\frac{3\varepsilon L}{2R^2}\right)$ will flow through the resistor ' R_0 ' after closing the switch-2.
4. An AC source of voltage varying as $V = 100 \sin(100t)$ volts is connected with resistance, capacitor and inductor as shown in the figure. Choose the **CORRECT** option(s).



- (A) Voltage across capacitor at time $t = \pi/4$ sec is 40 V.
- (B) At $t = 0$ inductor will gain energy at rate 16 J/s
- (C) At $t = 0$ capacitor will loose energy at rate 32 J/s
- (D) Power dissipated in circuit at $t = 0$ is 16 J/s.

Space for Rough Work

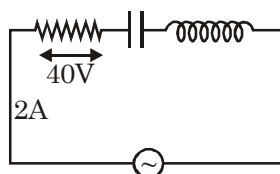
5. Assume that the Thompson's model incorporates Bohr's theory. In other words, in a hydrogen atom the electron moves inside a spherical volume of positive charge of the nucleus. The electric field in this spherically symmetric nucleus will provide the necessary centripetal force for revolution of electron. Assume that the angular momentum of the electron about the centre of nucleus is quantized. If n is the quantum number of a state,
- (A) Energy in n^{th} state is proportional to n .
 - (B) Speed in n^{th} state is proportional to $\sqrt{n}$.
 - (C) Radius of n^{th} state is proportional to n^2 .
 - (D) Magnetic field at the centre due to the motion of electron in n^{th} state is proportional to $n^{-1/2}$.
6. In an X ray tube the target is Tungsten ($\phi = 4.50$ eV) having atomic number 74. The voltage applied across the tube is 4 kV. Choose the **CORRECT** statement(s)
- (A) The minimum wavelength of the x rays emitted by the target is approximately 0.31 nm.
 - (B) The wavelength of K_{α} x rays is $\frac{4}{15987R}$ Where R is Rydberg's constant.
 - (C) If the work function of target is taken into account, the minimum wavelength of x rays emitted by the target increases slightly.
 - (D) If the work function of target is taken into account, the minimum wavelength of x rays emitted by the target decreases slightly.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

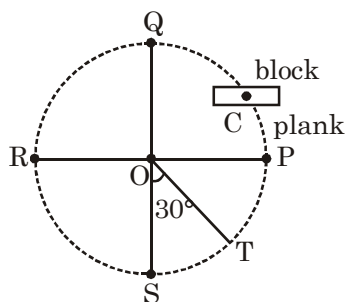
7. In the circuit shown in figure, match the following two columns. In Column II, quantities are given in SI units ($X_C = 30\Omega$, $X_L = 15\Omega$)



List-I	List-II
(P) Value of resistance R	(1) 60
(Q) Potential difference across capacitor	(2) 20
(R) Potential difference across inductor	(3) 30
(S) Applied potential difference	(4) 50
(A) (P) $\rightarrow$ (2) ; (Q) $\rightarrow$ (1) ; (R) $\rightarrow$ (3) ; (S) $\rightarrow$ (4)	
(B) (P) $\rightarrow$ (3) ; (Q) $\rightarrow$ (4) ; (R) $\rightarrow$ (2) ; (S) $\rightarrow$ (1)	
(C) (P) $\rightarrow$ (4) ; (Q) $\rightarrow$ (3) ; (R) $\rightarrow$ (1) ; (S) $\rightarrow$ (2)	
(D) (P) $\rightarrow$ (3) ; (Q) $\rightarrow$ (1) ; (R) $\rightarrow$ (4) ; (S) $\rightarrow$ (2)	

Space for Rough Work

8. A small block lies on a rough horizontal platform above its centre C as shown in figure. The plank is moved in vertical plane such that it always remains horizontal and its centre C moves in a vertical circle of centre O with the constant angular velocity ω . There is no relative motion between block and the plank and the block does not lose contact with the plank anywhere. P, Q, R and S, T are points on circular trajectory of centre C of platform. P and R lie on same horizontal level as O. Q is the highest point on the circle and S is the lowest point on the shown circle. Match the statements in List-I with points in List-II.



List-I

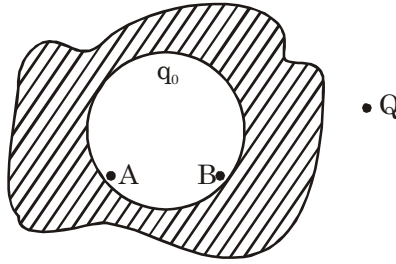
- (P) Magnitude of frictional force on block is maximum
 (Q) Magnitude of normal reaction on block is equal to mg
 (R) Magnitude of frictional force is zero
 (S) Net contact force on the block is directed towards the centre
 (A) (P) $\rightarrow$ (1,3) ; (Q) $\rightarrow$ (2,4) ; (R) $\rightarrow$ (1,3) ; (S) $\rightarrow$ (5)
 (B) (P) $\rightarrow$ (5) ; (Q) $\rightarrow$ (1,2,3) ; (R) $\rightarrow$ (2,4) ; (S) $\rightarrow$ (4,5)
 (C) (P) $\rightarrow$ (5) ; (Q) $\rightarrow$ (1,3) ; (R) $\rightarrow$ (2,4) ; (S) $\rightarrow$ (1,2,3,4,5)
 (D) (P) $\rightarrow$ (1,3) ; (Q) $\rightarrow$ (1,3) ; (R) $\rightarrow$ (2,4) ; (S) $\rightarrow$ (4)

List-II

- (1) When block is at position P
 (2) When block is at position Q
 (3) When block is at position R
 (4) When block is at position S
 (5) When block is at position T

Space for Rough Work

9. A point charge q_0 is placed inside a spherical cavity of a conductor. Another point charge Q is placed outside the conductor as shown.



List-I gives assertion statements and List-II gives its reason statements. Match the assertions with proper reason, (at v_∞ = potential at infinite = zero)

List-I

- (P) Total charge induced on cavity wall is equal and opposite to q_0 .
- (Q) As the point charge Q placed outside is pushed away from conductor, the potential difference ($V_A - V_B$) between two points A and B within cavity of sphere remains constant
- (R) Electric field due to induced charges on the inner surface of cavity at a point outside the conductor is non zero
- (S) Electric potential at a point outside the conductor due to induced charges on the inner surface of cavity is non-zero.

List-II

- (1) if cavity is surrounded by a gaussian surface, where all parts of gaussian surface are located inside conductor then.
 $\oint \vec{E} d\vec{A} = 0$. Hence, total charge enclosed by cavity = 0
- (2) The resultant electric field due to charges on outer surface of conductor and outside the conductor is zero at all points inside the conductor
- (3) The resultant electric field due to charges induced on inner surface of conductor cavity and charge q_0 placed inside cavity is zero at all points outside cavity
- (4) Net electric potential due to charges induced on inner Surface of cavity and charge q_0 placed inside cavity is zero at all points outside cavity
- (5) Net electric field inside cavity is zero everywhere

- (A) (P) $\rightarrow$ (1,2) ; (Q) $\rightarrow$ (2,4) ; (R) $\rightarrow$ (3,4) ; (S) $\rightarrow$ (2)
- (B) (P) $\rightarrow$ (2) ; (Q) $\rightarrow$ (4) ; (R) $\rightarrow$ (3) ; (S) $\rightarrow$ (1)
- (C) (P) $\rightarrow$ (1) ; (Q) $\rightarrow$ (2) ; (R) $\rightarrow$ (3) ; (S) $\rightarrow$ (3, 4)
- (D) (P) $\rightarrow$ (1,2) ; (Q) $\rightarrow$ (2,3) ; (R) $\rightarrow$ (1,3) ; (S) $\rightarrow$ (2,4)

Space for Rough Work

10. In the following question, in the List-I. we have listed certain situations and action in that situation. Match it with the outcome in List-II. In all cases consider angular momentum about centre.

List-I			List-II	
	Situation	Action		Effect
(P)	A satellite is revolving in a near earth circular orbit.	It enters a region of small atmospheric resistance.	(1)	The kinetic energy increases.
(Q)	An electron in H-atom revolving in ground state.	It absorbs a photon of energy 10.2 eV	(2)	The angular momentum increases.
(R)	An electron revolving in a circular path in a uniform magnetic field	The magnetic field is increased at a constant rate.	(3)	The kinetic energy decreases.
(S)	A sphere is kept on smooth ground. It is moving with velocity v_0 without rotation	It enters rough horizontal surface.	(4)	The angular momentum decreases
			(5)	Potential energy decreases.

- (A) (P) $\rightarrow$ (1, 4, 5) ; (Q) $\rightarrow$ (2, 3) ; (R) $\rightarrow$ (1, 2) ; (S) $\rightarrow$ (2, 3)
 (B) (P) $\rightarrow$ (3,4) ; (Q) $\rightarrow$ (1,2,5) ; (R) $\rightarrow$ (3,4) ; (S) $\rightarrow$ (2,3)
 (C) (P) $\rightarrow$ (3,4) ; (Q) $\rightarrow$ (1,5) ; (R) $\rightarrow$ (1,4) ; (S) $\rightarrow$ (3,4)
 (D) (P) $\rightarrow$ (4,5) ; (Q) $\rightarrow$ (2,3) ; (R) $\rightarrow$ (3,4) ; (S) $\rightarrow$ (2,3)

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

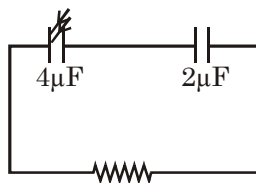
-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A 100 eV electron collides with a stationary He-ion (He^+) in its ground state, and excites it, to a higher level. After collision, He^+ ion emits two photons in succession with wavelength $1085\text{\AA}$ and $304\text{\AA}$. If the energy of electron after collision is $x\text{eV}$. Fill x in OMR sheet. (Assume $hc = 12400\text{ eV\AA}$)
2. A μ -meson (charge = $-e$, mass = $208 m_e$) moves in a circular orbit around a heavy nucleus having charge $+3e$. Find the quantum state n for which the radius of the orbit is same as that of first Bohr orbit for hydrogen atom.

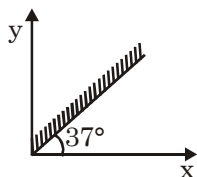
Space for Rough Work

3. Two capacitors of capacitance $4\mu\text{F}$ and $2\mu\text{F}$ are connected in series as shown.



The plates of both the capacitors are initially uncharged. They are made of metal of work function 2 eV . Suddenly we switch on a light source which emits monochromatic radiation of wavelength 400 nm . The power of the incident beam is 10 W . Assume that the photoelectric efficiency is 100% . What is the charge (in μC) on $2\mu\text{F}$ capacitor in steady state ? Assume that the resistance of resistor is 2Ω ? (Assume $hc = 12400\text{ eV}$)

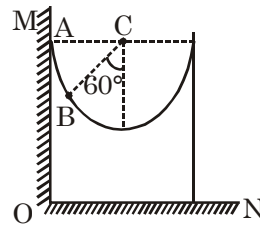
4. A light wave described by the equation $\vec{E} = \vec{E}_0 \sin\left(\frac{2\pi}{3} \times 10^7 \left(-\frac{4x}{5} + \frac{3y}{5}\right) - \omega t\right)$ is incident on a perfectly reflecting mirror surface whose area is 1 m^2 . The mirror is oriented at an angle of 37° to the x axis as shown. What is the pressure (in Pa) experienced by the mirror ? Assume that the light beam has an intensity of 300 MW/m^2 .



5. An electron of kinetic energy $K\text{ eV}$ strikes a hydrogen atom in ground state. Assume that the mass of the hydrogen atom is so large that it does not move at all. The electron is just able to excite the hydrogen atom to a higher excited state where the kinetic energy of the electron in the orbit is $K/3$. Find the value of K .

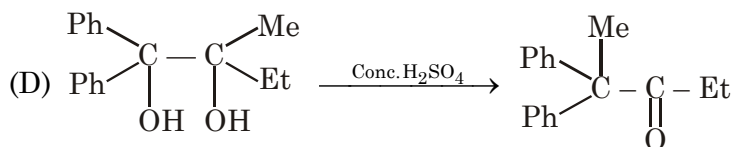
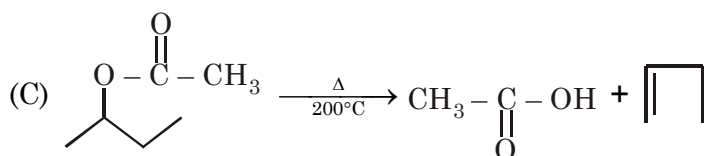
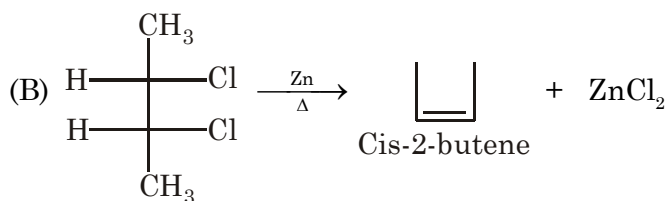
Space for Rough Work

6. The volume charge density of a solid sphere of radius a is directly proportional to the depth below the surface. The depth at which the electric field is maximum is a/n . Fill n in OMR sheet.
7. A train of mass 20000 kg is travelling with uniform velocity on a level track. Resistance to motion is proportional to mass and pull of the engine is constant. The last box of mass 4000 kg gets uncoupled. The driver discovers it after travelling a distance of 3.2 km and immediately shuts off engine but does not apply brakes. When both the parts come to rest the distance between them is x km. Then $x = ?$
8. A ball of mass 1 kg is released from position A inside a wedge with a hemispherical cut of radius 0.5 m as shown in the figure. Find the force (in N) exerted by the vertical wall OM on wedge, when the ball is in position B. (neglect friction everywhere). (Assume $\sqrt{3} = 1.732$)



Space for Rough Work

3. 250 ml of 1M I_2 is separately reacted with 0.5M Cu_2S solution, 0.5M CuS solution and 0.5 M $S_2O_3^{2-}$ solution causing production of Cu^{+2} and SO_4^{2-} in the first two and $S_4O_6^{2-}$ in the last case along with iodide ions. Which of the following option is/are correct assuming 100% completion of the reaction. ($I = 127$)
- (A) 100 ml of Cu_2S solution will be consumed.
- (B) 125 ml of CuS solution will be consumed.
- (C) 500 ml of $S_2O_3^{2-}$ solution will be consumed.
- (D) Equivalent weight of I_2 in each of the reaction will be 127.
4. Select the **CORRECT** order against the mentioned properties -
- (A) $K_4[Fe(CN)_6] > K_3[Fe(CN)_6]$ (EAN)
- (B) $[Mn(CO)_6]^+ > [Cr(CO)_6]^- > V(CO)_6^-$ (C – O bond order)
- (C) $\sigma - C_5H_5^- > \pi - C_2H_4 > \pi - C_6H_6$ (hapticity of ligand)
- (D) $imda^{2-} > nta^{3-} > ox^{2-}$ (denticity of ligand)
5. Which of the following option is / are incorrect?
- (A) Tertiary butyl alcohol gives immediate turbidity on reaction with Lucas reagent.



Space for Rough Work

6. Select the **CORRECT** statement for the same mole of gas -
- (A) Heat absorbed in isothermal reversible expansion for vanderwaal gas is more than ideal gas for same volume change.
 - (B) Heat liberated in isothermal reversible compression for vanderwaal gas is more than ideal gas for same volume change.
 - (C) Internal energy of vanderwaal gas increases during isothermal reversible expansion
 - (D) Internal energy of vanderwaal gas increases during isothermal reversible compression
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
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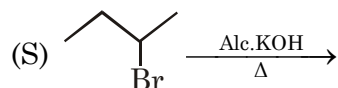
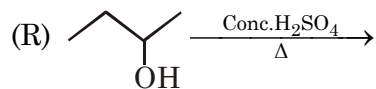
7.	List-I	List-II
(P)	C_2	(1) Same bond order as that of CN^-
(Q)	B_2	(2) Bond order of π bond is more than σ bond (whenever ' σ ' bond(s) is/are also present)
(R)	O_2	(3) Ions obtain by adding and removing one electron, shows same magnetic nature.
(S)	N_2	(4) Ionisation energy is lower as compare to corresponding atomic form
		(5) sp mixing is possible

The **CORRECT** option is

- (A) P→2,3,4,5; Q→2,3,4,5; R→3; S→1,3,4,5
 (B) P→1,2,3; Q→ 1,2,3; R→4; S→3,5
 (C) P→2,3,5; Q→2,3,5,; R→3,4; S→1,2,3,5
 (D) P→3,5; Q→3,5; R→3,4; S→1,2,3,5

Space for Rough Work

8. List-I



List-II

(1) Unimolecular elimination

(2) Bimolecular elimination

(3) Racemic mixture formed

(4) Major product shows stereoisomerism

(5) Electrophilic addition

The correct option is -

(A) P $\rightarrow$ 1,3,4,5; Q $\rightarrow$ 3,4,5; R $\rightarrow$ 1,4; S $\rightarrow$ 2,4

(B) P $\rightarrow$ 3,4,5; Q $\rightarrow$ 3,4,5; R $\rightarrow$ 1,4; S $\rightarrow$ 2,3,4

(C) P $\rightarrow$ 3,4,5; Q $\rightarrow$ 3,4,5; R $\rightarrow$ 1,4; S $\rightarrow$ 2,4

(D) P $\rightarrow$ 4,5; Q $\rightarrow$ 3,4,5; R $\rightarrow$ 1,4; S $\rightarrow$ 2,4

Space for Rough Work

9. List-I

List-II

- | | |
|---|---|
| (P) Cation in ccp and anion in alternate tetrahedral void | (1) Ratio of number of cation to anion in one unit cell 1 : 1 |
| (Q) Cation in simple cubic and anion in the body centre | (2) Ratio of coordination no. of cation to anion is 1 : 1 |
| (R) Anions in CCP and cation in all tetrahedral voids | (3) Ratio of coordination number of cation to anion is not 1 |
| (S) Cation in ccp and anions in all is octahedral voids | (4) Number of next neighbours of ion greater than 10 |

The correct option is

- | | |
|----------------------------------|--------------------------------|
| (A) P→1,2,4; Q→1,2; R→4; S→1,2,4 | (B) P→4; Q→3; R→1,2,4; S→1,2,3 |
| (C) P→1,2,4; Q→2; R→1,2,4; S→4 | (D) P→4; Q→1,2,4; R→2; S→1,2,4 |

10. List-I

List-II

- | | |
|---|--|
| (P) $[\text{Pt}(\text{NH}_2-\text{CH}(\text{CH}_3)-\text{CO}_2)_2]$ | (1) Central atom uses d_{z^2} orbital in hybridisation |
| (Q) $[\text{Cr}(\text{ox})_3]^{3-}$ | (2) Shows optical isomerism |
| (R) $[\text{CoF}_6]^{3-}$ | (3) Complex is inner orbital complex |
| (S) $[\text{Rh}(\text{gly})_3]$ | (4) Shows geometrical isomerism |
| | (5) Complex contains symmetrical bidentate ligand |

The correct option is

- | | |
|--|--|
| (A) P→1,2,3,5; Q→1,2,3,5; R→3; S→1,2,4 | (B) P→2,3,4; Q→1,2,3,4; R→1,3; S→2,3 |
| (C) P→1,2,3,4; Q→1,2,3,5; R→1; S→1,2,3,4 | (D) P→2,3,4; Q→1,2,3,5; R→1; S→1,2,3,4 |

Space for Rough Work

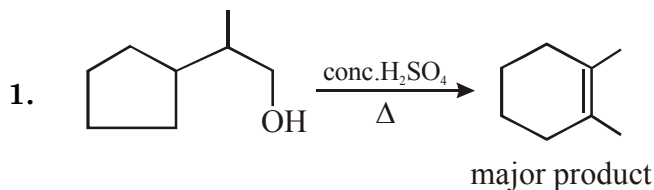
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+										-									
●	●	○	○	○	○	●	●	●	○	○	○								
①	①	①	①	①	①	①	①	①	①	①	①								
②	②	②	②	●	②	②	②	②	②	●	②								
③	③	③	③	③	③	③	③	③	③	③	③								
④	④	④	④	④	④	④	④	④	④	④	④								
⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	●	⑤	⑤								
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥								
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦								
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧								
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨								

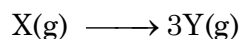
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.



Number of 1, 2 shifts involved in above reaction.

Space for Rough Work

2. Given :



$P_t = 400 \text{ mmHg}$: Total pressure after time 70 min

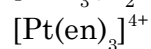
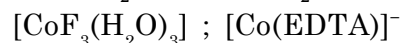
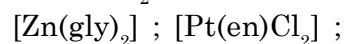
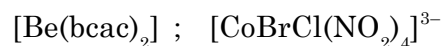
$P_\infty = 600 \text{ mmHg}$: Total pressure after completion

$$t_{0.75} = 2 \times t_{0.5}$$

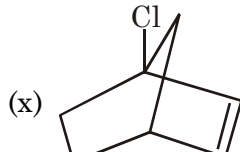
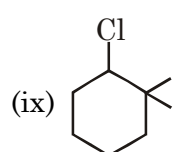
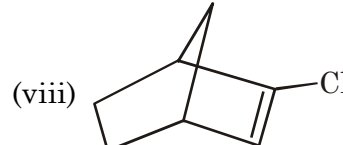
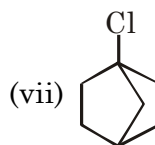
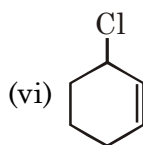
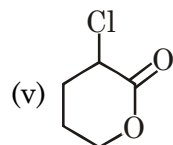
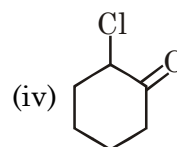
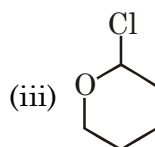
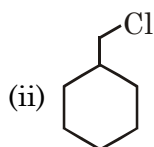
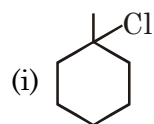
Calculate rate constant of appearance of Y in minute^{-1} -

given $\ln 2 = 0.7$

3. Find out the ratio of optically active to optically inactive compounds among the following



4. How many alkyl halide gives $\text{S}_{\text{N}}2$ reaction faster than chlorocyclohexane with NaOH?

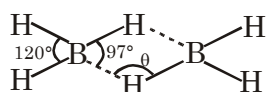


Space for Rough Work

5. A light source emits radiations of $\lambda = 310$ nm. Number of photons emitted by this source varies with time, as $N = 10^{18}t^2$. (t in sec). If this light source is used to emit photoelectrons from a metal of work function 3.1 eV. Find the time (in sec.) during which the total maximum possible kinetic energy of all the emitted photoelectrons from the beginning is 1296 J.

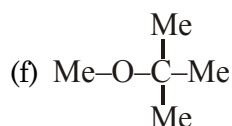
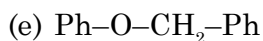
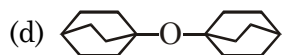
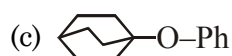
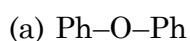
(Use $E(\text{eV}) = \frac{12400}{\lambda(\text{\AA})}$, $e = 1.6 \times 10^{-19} \text{ C}$).

6. For the dimer of BH_3 ,



Find out the value of θ .

7. Total number of ether which can not formed by Williamson ether synthesis.



8. 560 gm, 2m Urea solution when kept at temperature which is 5°C below its freezing point, then half of the solvent freezes out. Calculate the weight (in gm) of solvent freezes out if the same solution is placed at temperature 15°C below its freezing point

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f(x) = 2^x + 4^x + 8^x$ and $g(x)$ is inverse function of $f(x)$, then (where $[.]$ represents greatest integer function)-

(A) $g'(14) \ln 2 < 1$ (B) $[g'(14) \ln 2] = 0$ (C) $g'(14) \ln 2 > 1$ (D) $[g'(14) \ln 2] = 1$

2. Let $y = f(x)$ is a positive valued function which satisfies equation $\sqrt{y^2 + 2x} + \sqrt{y^2 - 2x} = 2x^2$,

then $\frac{dy}{dx}$ is equal to-

(A) $\frac{2x^3 - x^{-3}}{y}$ (B) $\frac{2x^4 - x^{-2}}{y}$ (C) $\frac{2x^4 - x^{-2}}{\sqrt{x^6 + 1}}$ (D) $\frac{2x^3 - x^{-3}}{\sqrt{1 + x^6}}$

Space for Rough Work

3. Let $f: \left[-\frac{1}{3}, 3\right] \rightarrow \mathbb{R}$ and $g: \left[-\frac{1}{3}, 3\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 4]$ and $g(x) = |x-2| f(x) + |3x-5| f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

- (A) f is discontinuous exactly at eight points in $\left[-\frac{1}{3}, 3\right]$
- (B) f is discontinuous exactly at nine points in $\left[-\frac{1}{3}, 3\right]$
- (C) g is NOT differentiable exactly at ten points in $\left(-\frac{1}{3}, 3\right)$
- (D) g is NOT differentiable exactly at nine points in $\left(-\frac{1}{3}, 3\right)$

Space for Rough Work

4. Consider $f(x) = (x-1)\tan^{-1}x - \frac{1}{2}\ln(1+x^2)$, $x \geq 1$. Identify which of the following statement(s) is(are) correct ?
- (A) $\lim_{x \rightarrow \infty} f'(x) = \frac{\pi}{2}$ (B) $f'(x)$ is increasing in $[1, \infty)$
- (C) $f(x+4) - f(x) > 2\pi \forall x \in [1, \infty)$ (D) $f(x+4) - f(x) < 2\pi \forall x \in [1, \infty)$
5. Consider $P(x) = 9x^8 - 18x^5 - 20x^3 + 15$, then $P(x) = 0$ has
- (A) atleast one real root in $[0, 3^{1/3}]$ (B) atleast one real root in $[5^{1/5}, 3^{1/3}]$
- (C) atleast one real root in $[0, 5^{1/5}]$ (D) no real root in $[0, 3^{1/3}]$
6. $\int \sqrt{1 + \operatorname{cosec} x} dx$ equals - (where C is integration constant)
- (A) $2 \sin^{-1} \sqrt{\sin x} + C$ (B) $\sqrt{2} \cos^{-1} \sqrt{\cos x} + C$
- (C) $C - 2 \sin^{-1}(1 - 2 \sin x)$ (D) $\cos^{-1}(1 - 2 \sin x) + C$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I

List-II

- | | |
|---|---|
| (P) $f(x)$ is increasing function in | (1) $\left(0, \frac{3}{2}\right)$ |
| (Q) $x = C$ is point of inflection for curve $y = f(x)$, then C lies in | (2) $\left(\frac{3}{2}, 4\right)$ |
| (R) Let N denotes number of real root(s) of equation $f(x) = 0$, then N lies in | (3) (5,7) |
| (S) Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is sum of all possible values of a , then k lies in | (4) (8,10)
(5) (9,11) |
| (A) $P \rightarrow 4, 5$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$ | (B) $P \rightarrow 4, 5$; $Q \rightarrow 2$; $R \rightarrow 2$; $S \rightarrow 4, 5$ |
| (C) $P \rightarrow 5$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 4, 5$ | (D) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 5$ |

Space for Rough Work

- | 8. | List-I | List-II |
|-----|---|--|
| (P) | If $y = \ln(x^2) $ then $y'\left(-\frac{1}{2}\right)$ is equal to | (1) 4 |
| (Q) | If $f(x) = \frac{1}{2}x^3 + \tan^{-1}x - \frac{\pi}{4}$ and $g(x) = f^{-1}(x)$,
then $\left 16g''\left(\frac{1}{2}\right)\right $ is equal to | (2) 5
(3) 2 |
| (R) | If $f(x) = x^{x^x}$, then $f''(1)$ is equal to | (4) 1 |
| (S) | If $\ln((e-1)e^{xy} + x^2) = x^2 + y^2$, then $\left.\frac{dy}{dx}\right _{(1,0)}$ is | (5) Harmonic mean of 3 and $\frac{3}{2}$ |
| | (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3,5$ | (B) $P \rightarrow 1,5; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$ |
| | (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3,5; S \rightarrow 3,5$ | (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3,5; S \rightarrow 3,5$ |
-
- | 9. | List-I | List-II |
|-----|--|--|
| (P) | Value of λ for which the $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 12\lambda x$ has one local maxima and one local minima in R , can be | (1) 0 |
| (Q) | If $f(x) = x^{\frac{1}{\ln x}}$ then $f'(x) = 0$ for x equals to | (2) 1 |
| (R) | Cosine of the angle of intersection of curves $f(x) = 2^x \ln x$ and $g(x) = x^{2x} - 1$ is | (3) 2 |
| (S) | Consider curve $y^3 - 3xy + 2 = 0$ then total no of horizontal & vertical tangent is | (4) 3
(5) e |
| | (A) $P \rightarrow 1,2,4,5; Q \rightarrow 3,4,5; R \rightarrow 2; S \rightarrow 3$ | (B) $P \rightarrow 2,4,5; Q \rightarrow 2,3,4,5; R \rightarrow 2; S \rightarrow 3$ |
| | (C) $P \rightarrow 1,2,4,5; Q \rightarrow 3,4,5; R \rightarrow 3; S \rightarrow 2$ | (D) $P \rightarrow 1,2,4,5; Q \rightarrow 2,3,4,5; R \rightarrow 3; S \rightarrow 2$ |

Space for Rough Work

10. Let $f(x) = \begin{cases} x+a & , x < 0 \\ \sqrt{x^2-2x+1} & , x \geq 0 \end{cases}$

and $g(x) = \begin{cases} x+1 & , x < 0 \\ x^2 + 2(b-x)+1 & , x \geq 0 \end{cases}$

a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all real x

List-I

List-II

(P) Number of possible values of b is

(1) 0

(Q) Sum of all possible values of a is

(2) 1

(R) Value of $\frac{d(g(f(x)))}{dx}$ at $x = 4$ is

(3) 2

(S) If $f(x)$ is non differentiable at k points, then the value of k is

(4) 4

(5) greater than 4

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If a cuboid, length of whose one edge is $\sqrt{2}$ units, is inscribed in a sphere of radius $\sqrt{2}$ units, then maximum possible volume of cuboid is : (in cubic unit)
2. If $\int \frac{e^x(x^2 - x + 1)}{(x^2 + 1)\sqrt{x^2 + 1}} dx = e^x \cdot f(x) + C$, where C is constant of integration, then number of integers in the range of $f(x)$ is
3. Let $f(x)$ be a twice differentiable function satisfying $xf(x^2) + (x^2 + 1)f(x) = x$, $\forall x \in \mathbb{R}$ then the value of $|f''(1)|$ is
4. $\lim_{x \rightarrow 0^+} (x \cos x)^x + (x \sin x)^{1/x}$ is

Space for Rough Work

5. Let $4\left(\frac{ay}{\cos \alpha} - \frac{bx}{\sin \alpha}\right) = 3(a+b)$, $ab \neq 0$ is normal to the curve $x = 3\cos\theta - \cos^3\theta$, $y = 3\sin\theta - \sin^3\theta$ at $\theta = \alpha$, then number of possible values of α , $\alpha \in (0, \pi) - \left\{\frac{\pi}{2}\right\}$ is
6. Number of point(s) in $[0, 9]$, where $f(x) = \{x\} \left| \sin \frac{\pi x}{2} \right| - \left\{ \frac{x}{2} \right\}$ is discontinuous, is
(where $\{.\}$ denotes fractional part function)
7. The ordinate of a point describing the circle $x^2 + y^2 = 25$ decreases at the rate of 1 cm/sec, then the rate of change (in cm/sec) of abscissa of the point when ordinate equal to 3cm, is-
(Given $x > 0$, $y > 0$)
8. The maximum value of $f(x) = \frac{2x^2 + 7x + 3}{2x^2 + x + 1}$, is

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QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator